

Question number	Answer	Mark
8(a)	B species richness of all the species within a habitat	(1)

Question number	Answer	Mark
8(b)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. Indicative content • good diversity for birds in all sections except Section 4, which decreases to moderate • good diversity for birds is very similar in Sections 1, 2 and 3 • moderate diversity for fish in Section 1, which increases in Section 2 but then decreases again in Sections 3 and 4 • moderate diversity is very similar in the three sections • good diversity for aquatic plants in Sections 1 and 2, which has decreased by Section 4 • good biodiversity for mussels in stretch 1, decreasing to moderate biodiversity in stretch 2 and poor diversity in stretch 4 • no overall trends as you move down the river	(6)

Level	Marks	Descriptor
	0	No awardable content.
1	1-2	Demonstrates isolated elements of biological knowledge related to the given context with generalised comments made. The description will contain basic information with some attempt made to link knowledge and understanding to the given context.
2	3-4	Demonstrates adequate knowledge by selecting and applying some relevant biological facts/concepts to provide the description being presented. The description shows some linkages and lines of reasoning with some structure.
3	5-6	Demonstrates comprehensive knowledge by selecting and applying relevant knowledge of biological facts/concepts to provide the description being presented. The description is clear, coherent and logically structured.

Question number	Answer	Mark
8(c)	An answer that includes the following points: • there are more species of fish than birds (1) • therefore a value of 2.5 has a relatively lower biodiversity than a value of 1.6 (1)	(2)

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8(d)(i)	<table><tr><th>Species of aquatic plant</th><th>Number of aquatic plants counted</th><th>(n-1)</th><th>n(n-1)</th></tr><tr><td>Coontail</td><td>8</td><td>7</td><td>56</td></tr><tr><td>Tape grass</td><td>6</td><td>5</td><td>30</td></tr><tr><td>Common waterweed</td><td>3</td><td>2</td><td>6</td></tr><tr><td>Northern water milfoil</td><td>2</td><td>1</td><td>2</td></tr><tr><td>Star duckweed</td><td>9</td><td>8</td><td>72</td></tr><tr><td>White water lily</td><td>2</td><td>1</td><td>2</td></tr><tr><td>Water star-grass</td><td>2</td><td>1</td><td>2</td></tr><tr><td>Eurasian water milfoil</td><td>6</td><td>5</td><td>30</td></tr><tr><td>Curly pondweed</td><td>5</td><td>4</td><td>20</td></tr><tr><td>European frogbit</td><td>2</td><td>1</td><td>2</td></tr><tr><td>Flowering rush</td><td>3</td><td>2</td><td>6</td></tr></table>	Species of aquatic plant	Number of aquatic plants counted	(n-1)	n(n-1)	Coontail	8	7	56	Tape grass	6	5	30	Common waterweed	3	2	6	Northern water milfoil	2	1	2	Star duckweed	9	8	72	White water lily	2	1	2	Water star-grass	2	1	2	Eurasian water milfoil	6	5	30	Curly pondweed	5	4	20	European frogbit	2	1	2	Flowering rush	3	2	6	(1)
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8(d)(ii)	• $\sum n(n-1)$ is 228 (1) • value for $N(N-1)$ is $(11 \times 10) = 110$ (1) • diversity index is 0.48 (1)	(3)